

MealEase Planner Pseudocode

None

Start

Display "Welcome to MealEase Planner"

Call getUserInformation()

Call getWeeklyBudget()

Call createMealPlan()

Call createGroceryList()

Set groceryTotal = calculateGroceryTotal()

Call compareBudget(groceryTotal)

Call displayMealPlan()

Call displayGroceryList(groceryTotal)

Stop

Get User Information

None

Module getUserInformation()

String userName

Integer familySize

Display "Enter your name:"

Input userName

Display "Enter the number of people in your family:"

Input familySize

While familySize <= 0

```
        Display "Family size must be greater than zero."
        Display "Enter the number of people in your family:"
        Input familySize
    End While
```

```
End Module
```

Get Weekly Budget

None

```
Module getWeeklyBudget()

    Real weeklyBudget

    Display "Enter your weekly grocery budget:"
    Input weeklyBudget

    While weeklyBudget <= 0
        Display "Invalid budget."
        Display "Enter a budget greater than zero:"
        Input weeklyBudget
    End While
```

```
End Module
```

Create Weekly Meal Plan

None

```
Module createMealPlan()

    Integer day
    String dayNames[7] = "Monday", "Tuesday", "Wednesday",
                        "Thursday", "Friday", "Saturday",
                        "Sunday"
```

```

String breakfast[7]
String lunch[7]
String dinner[7]
String snack[7]

For day = 0 To 6

    Display "Enter breakfast for ", dayNames[day], ":"
    Input breakfast[day]

    Display "Enter lunch for ", dayNames[day], ":"
    Input lunch[day]

    Display "Enter dinner for ", dayNames[day], ":"
    Input dinner[day]

    Display "Enter snack for ", dayNames[day], ":"
    Input snack[day]

End For

End Module

```

Create Grocery List

None

```

Module createGroceryList()

    Integer index
    Integer numberOfItems
    String groceryItem[]
    Real itemPrice[]

    Display "Enter the number of grocery items:"
    Input numberOfItems

```

```

While numberOfItems <= 0
    Display "Enter at least one grocery item:"
    Input numberOfItems
End While

For index = 0 To numberOfItems - 1

    Display "Enter grocery item ", index + 1, ":"
    Input groceryItem[index]

    Display "Enter the price of ", groceryItem[index], ":"
    Input itemPrice[index]

    While itemPrice[index] < 0
        Display "Price cannot be negative."
        Display "Enter the price again:"
        Input itemPrice[index]
    End While

End For

End Module

```

Calculate Grocery Total

None

```

Function Real calculateGroceryTotal()

    Integer index
    Real groceryTotal = 0

    For index = 0 To numberOfItems - 1
        Set groceryTotal = groceryTotal + itemPrice[index]
    End For

    Return groceryTotal

```

End Function

Compare Grocery Total With Budget

None

```
Module compareBudget(Real groceryTotal)

    Real remainingMoney
    Real overAmount

    If groceryTotal <= weeklyBudget Then

        Set remainingMoney = weeklyBudget - groceryTotal

        Display "Your meal plan is within budget."
        Display "Weekly budget: $", weeklyBudget
        Display "Grocery total: $", groceryTotal
        Display "Money remaining: $", remainingMoney

    Else

        Set overAmount = groceryTotal - weeklyBudget

        Display "Your meal plan is over budget."
        Display "Weekly budget: $", weeklyBudget
        Display "Grocery total: $", groceryTotal
        Display "Amount over budget: $", overAmount

    End If

End Module
```

Display Weekly Meal Plan

None

```
Module displayMealPlan()  
  
    Integer day  
  
    Display "MealEase Weekly Meal Plan"  
    Display "-----"  
  
    For day = 0 To 6  
  
        Display dayNames[day]  
        Display "Breakfast: ", breakfast[day]  
        Display "Lunch: ", lunch[day]  
        Display "Dinner: ", dinner[day]  
        Display "Snack: ", snack[day]  
        Display "-----"  
  
    End For  
  
End Module
```

Display Grocery List

None

```
Module displayGroceryList(Real groceryTotal)  
  
    Integer index  
  
    Display "Grocery Shopping List"  
    Display "-----"  
  
    For index = 0 To numberOfItems - 1  
        Display groceryItem[index], " - $", itemPrice[index]  
    End For  
  
    Display "Estimated Grocery Total: $", groceryTotal
```

End Module

Complete Program Flow

None

Start

Display welcome message

Input user name

Input family size

Validate family size

Input weekly grocery budget

Validate grocery budget

For each day Monday through Sunday

Input breakfast

Input lunch

Input dinner

Input snack

End For

Input number of grocery items

For each grocery item

Input grocery item name

Input grocery item price

End For

Calculate grocery total

If grocery total is less than or equal to weekly budget Then

Calculate remaining money

Display "Within Budget"

Else

Calculate amount over budget

Display "Over Budget"

End If

```
    Display weekly meal plan
    Display grocery list
    Display grocery total
Stop
```